ANSWER KEY

Quiz: Lesson 1 — Data Management and Analysis in Excel

Course: Productivity Tools for Technical Reporting

Questions

Multiple Choice

****Q1.** You are organizing a 30-day OrbitCo incident analysis workbook. Which of the following is the most defensible worksheet structure for a workbook a reviewer may audit months later?**

☐ a) One sheet with raw data, cleaned data, and charts arranged in side-by-side regions

☐ b) One sheet per analysis question, with each sheet holding its own raw-and-cleaned copy

■ **c) Separate sheets for raw data, processed data, and outputs (Key Performance Indicators and charts), so the raw sheet remains untouched as evidence**

☐ d) A single sheet with color-coding to distinguish raw rows from cleaned rows

Correct: c

> Layered worksheets preserve the raw → processed → outputs lineage. If a reviewer challenges a number, you can walk it back to the original export and prove it. Option (a) risks overwriting raw data during cleanup. Option (b) duplicates data and drifts across copies. Option (d) makes color the data — but color is a render, invisible to formulas and to anyone who prints the file in black and white.

Q2. An OrbitCo ticket export contains nineteen rows where `Severity` is blank because the support agent closed the ticket before triage. For a 30-day operational review, what is the most defensible cleanup choice?

- [ ] a) Delete the nineteen rows so severity counts add up cleanly
- [ ] b) Fill the blanks with "Medium" because it is the most common severity in the dataset
- [x] **c) Add a `SeverityIsMissing` flag column using `=ISBLANK([@Severity])`, keep the rows in the Table, and report the missing-severity count as its own line in the Key Performance Indicator summary**
- [ ] d) Hide the rows with an AutoFilter so they do not appear in reports

Correct: c

> Flagging preserves the accurate total ticket count (413) and documents the data-quality issue honestly rather than hiding it. Deleting silently drops the count to 394, and leadership will ask where the other nineteen went. Imputing invents a severity that was never assigned. Hiding just defers the same problem to the next person to open the workbook.

Q3. An OrbitCo monitoring export shows 43 total minutes of downtime across a 30-day window. Which Excel formula correctly computes uptime percentage when the cleaned downtime values are stored in a Table named `Uptime` with a column `DowntimeMinutes`?

- [ ] a) `=1 - 43/30`
- [ ] b) `=1 - SUM(Uptime[DowntimeMinutes])/(30*24)`
- [x] **c) `=1 - SUM(Uptime[DowntimeMinutes])/(30*24*60)`**
- [ ] d) `=100 - SUM(Uptime[DowntimeMinutes])`

Correct: c

> Uptime percentage is one minus the ratio of downtime to total time in the window. The total minutes in a 30-day window are `30 × 24 × 60 = 43,200`. Option (a) divides by days, giving a nonsensical fraction. Option (b) divides by hours, undercounting the denominator by a factor of 60 and exaggerating the downtime. Option (d) mixes units (subtracting minutes from the number 100) and produces a negative number of questionable meaning.

Multiple Select

****Q4.** Which of the following are features you gain by converting a plain data range into an Excel Table (Ctrl+T)? *(Select all that apply.)***

- [x] **a) Auto-filter dropdowns appear automatically in the header row**
- [x] **b) Formulas entered in one cell of a Table column auto-fill down the entire column**
- [ ] c) Cells become protected from accidental editing
- [x] **d) Structured references like `Tickets[Severity]` become valid anywhere in the workbook**
- [x] **e) New rows added at the bottom inherit formulas, formatting, and data validation**

Correct: a, b, d, e

> Tables deliver (a) auto-filter, (b) formula auto-fill on new rows, (d) structured references, and (e) inheritance of formulas, formatting, and data validation. Option (c) is false — cells in a Table are not protected from editing. Protection is a separate feature (Review → Protect Sheet) and is not conferred by Ctrl+T. Understanding what a Table does and does not provide prevents false confidence in workbook integrity.

True/False

****Q5.** Writing `=COUNTIF(Tickets[Severity], "High")` is more reliable than writing `=COUNTIF(G2:G421, "High")` in an operations workbook that receives new ticket rows every week.**

Correct: True

> The structured reference `Tickets[Severity]` automatically expands as new rows are appended to the Table, so the formula keeps counting every High-severity ticket without any manual edit. The hardcoded range `G2:G421` silently stops counting the moment a new row lands on row 422 — producing a wrong answer that looks right because the formula returns a number. This is the central reliability argument for Tables and structured references, and it is why the On-the-Job Learning (OJL) 5d competency tag applies to this material.